PHISON

aiDAPTIV+

aiDAPTIVLink Specification (NXUN205.A1)

Version 1.0
Release date 2026/3/12

Introduction

aiDAPTIV: Integrated AI Computing Solution Overview

aiDAPTIV is a comprehensive AI computing solution designed to enable cost-effective **LLM training and inference** on local systems, including workstations, PCs, and edge devices.

By integrating **AI SSD- aiDAPTIVCache, and optimized middleware- aiDAPTIVLink**, aiDAPTIV extends GPU memory capacity by leveraging high-performance flash memory as a secondary memory tier. This allows organizations to execute and fine-tune massive AI models without the need for excessive, high-cost GPUs or High Bandwidth Memory (HBM), drastically lowering the hardware barrier for AI adoption.

aiDAPTIVLink 2 Series: Post-training & Fine-tuning

The **aiDAPTIVLink 2 Series** is a software solution specifically engineered for the **post-training (Fine-tuning)** of LLMs and VLMs. Its primary innovation lies in the **optimized allocation of resources** during the training phase.

By pairing with **Phison AI SSDs**, the software intelligently offloads training workloads to the SSD, streaming data to the VRAM only when active GPU computation is required. This architecture significantly alleviates the demand for high-capacity VRAM and enables **larger Batch Sizes** for increased throughput. Consequently, it minimizes the required GPU count and overall infrastructure costs while maintaining competitive training timelines.

aiDAPTIVLink 3 Series: Inference Acceleration

The **aiDAPTIVLink 3 Series** is an acceleration software purpose-built for AI model **inference**. When integrated with **Phison AI SSDs**, the software enables the offloading of **KV Cache** generated during the inference process onto the SSD.

In scenarios requiring data reuse, the system retrieves the KV Cache directly back to the VRAM, effectively **eliminating redundant computations**. This optimization significantly reduces the **Time To First Token (TTFT)**, delivering a faster, more responsive, and seamless inference experience.

COPYRIGHT © 2024 PHISON ELECTRONICS, ALL RIGHTS RESERVED.

Specification

Version	NXUN205.A1
Environment Requirement	OS: Ubuntu 24.04.3 Desktop Kernel: 6.14 GPU driver: NVIDIA driver version 580 installation NVIDIA Toolkit: CUDA 13.0 installation
aiDAPTIVCache	PASCARI AI100E (PCIe 4.0) PASCARI AI200E (PCIe 5.0)
Training Features	Customized training_dataloader/lr_scheduler Support CPU_Adam Optimizer Support Lora Finetune Support Full Parameter Finetune Support Offload GPU Memory to Phison SSD Support VLM Support Multi-node
Distributed GPU	Support 1, 2, 4, 8 GPUs
GPU AVL	NVIDIA H200 NVIDIA H100 NVIDIA RTX A6000 NVIDIA RTX 6000 Ada NVIDIA RTX 5000 Ada NVIDIA RTX 4000 Ada NVIDIA L40/L40S NVIDIA RTX 50 series NVIDIA RTX 40 series NVIDIA RTX Pro 6000
CPU AVL	Intel Xeon Gold 5320 Intel Xeon Gold 6330 Intel Xeon w5-3425 Intel Xeon Gold 6538 Y+ Intel Xeon Silver 4410Y Intel Xeon Silver 4410T Intel Xeon Silver 5315Y Intel i9-13900 Intel i9-12900E AMD Ryzen Threadripper 7980X 64-Cores AMD EPYC 7713P AMD EPYC 9174F

COPYRIGHT © 2024 PHISON ELECTRONICS, ALL RIGHTS RESERVED.

Distributed Nodes	Support 1 node/ Multi-node
Model Support List	Llama-3-Taiwan-70B-Instruct Llama_3.1_8B_Instruct Llama_3.1_70B_Instruct Llama_3.2_3B_Instruct Llama_3.2_11B_Vision_Instruct Llama_3.3_70B_Instruct Qwen2.5_7B_Instruct Qwen2.5_32B_Instruct Qwen2.5_72B_Instruct Qwen3-14B Qwen3_32B QwQ_32B DeepSeek_R1_Distill_Llama_70B DeepSeek_R1_Distill_Qwen_32B DeepSeek-R1-Distill-Qwen-7B gemma_3_27b_it Phi-4-multimodal-instruct Phi_4 gpt_oss_20b gpt_oss_120b Mistral_Small_3.1_24B_Instruct_2503 whisper_large_v3_turbo
Sequence Length	Default: 2K (Maximum value depends on each AI model)

COPYRIGHT © 2024 PHISON ELECTRONICS, ALL RIGHTS RESERVED.

PHISON | aiDAPTIV+